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The Palaeomagnetism of Some Cambrian and Ordovician Sediments from the Northern Territory, Australia

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Summary

Samples for palaeomagnetic investigation were collected from nine horizons in the Jinduckin Formation, of Early Ordovician age, and from four exposures of a red mudstone band within the Middle Cambrian Montejinni Limestone. The mudstone samples responded unsatisfactorily to both alternating field and thermal demagnetization procedures and their magnetization appears to originate from chemical action during weathering in recent times. In contrast, only one site from the Jinduckin Formation failed to respond to thermal cleaning, and a chemical demagnetization technique here was also unsuccessful. Thermally cleaned directions from seven of the other eight sites are assumed to yield the original direction of magnetization of the Jinduckin Formation ($D = 252°$, $I = -14°$, $\alpha_{95} = 20°$), and are used to derive the mean pole position for the period of its deposition (Long. = 29° E, Lat. = 16° S, $A_{95} = 17°$). This is the first result from a current palaeomagnetic study of the lower Palaeozoic red beds of northern Australia.

Introduction

Early palaeomagnetic studies of various Mesozoic rock formations from Africa (Gough, Opdyke & McElhinny 1964) and Australia (Irving, Robertson & Stott 1963) together with one from South America (Creer 1962) demonstrated the occurrence of relative movement between these land masses since Mesozoic times. The African and Australian results showed that during the Mesozoic era the pole remained stationary with respect to both continents so that, in effect, only one Mesozoic pole position was determined for each. Now a single pole position places a continent in latitude and azimuthal orientation but is insufficient to fix its longitude; therefore the configuration of the southern land masses prior to continental drift could not be uniquely determined from the Mesozoic data.

It is possible, however, to reconstruct without ambiguity the relative positions of two or more continents if they have remained in a fixed location with respect to each other during a period in which substantial polar wandering has occurred (Irving 1958; Graham, Helsley & Hales 1964). Results from South America (Creer 1965) and Africa (McElhinny *et al.* 1968) indicate that this condition was satisfied during the early and middle parts of the Palaeozoic era, and McElhinny (1967) has shown that the polar wander paths for the two continents are virtually coincident for most of the Palaeozoic if the land masses were adjacent at that time. To determine the relative position of Australia in such a reconstruction, conclusive palaeomagnetic results from the lower Palaeozoic are essential, but only preliminary measurements have so far been reported (Irving & Green 1958).

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A systematic investigation of lower Palaeozoic rocks from northern Australia is now in progress and the present paper reports the first results from this study. Measurements on the Montejinni Limestone and the Jinduckin Formation are described.

Geology

The Cambrian and Ordovician sediments of the Northern Territory (Fig. 1) are mostly covered by a thin veneer of Quaternary deposits and sampling is normally restricted to small, isolated exposures. In general, as shown by the stratigraphic column in Table 1, a well-dated sequence of marine Middle Cambrian to Lower Ordovician sediments disconformably overlies the Antrim Plateau Volcanics of Lower Cambrian age. The region has been tectonically quiet since Precambrian times (Randal & Brown 1967).

The Montejinni Limestone is comprised of two fossiliferous limestone units separated by a red-brown mudstone unit which appears to be the only palaeomagnetically promising material of the Middle Cambrian sequence. Unfortunately, this mudstone band is subject to preferential weathering and fresh exposures are

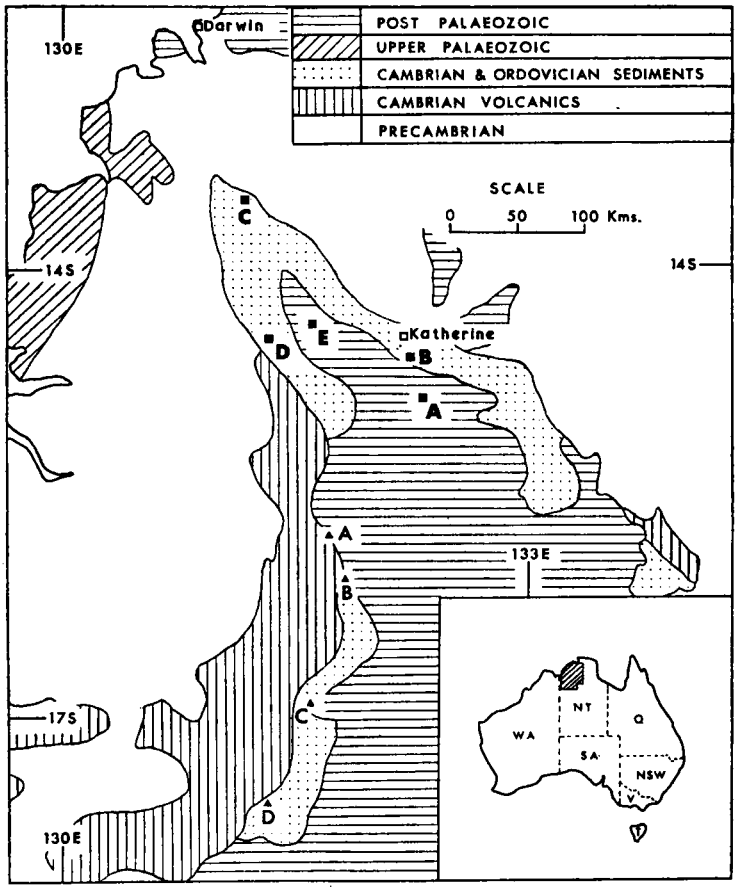


FIG. 1. Geological sketch map of part of the Northern Territory showing sampling sites in the Jinduckin Formation (squares) and the Montejinni Limestone (triangles).

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Table 1

Lower Palaeozoic stratigraphy within the Northern Territory

Lower Ordovician	Oolloo Limestone
	Jinduckin Formation
	? ?
	Manbulloo Limestone
Middle Cambrian	Merrina Beds
	Tindall Limestone
	Montejinni Limestone
	Disconformity
Lower Cambrian	Antrim Plateau Volcanics

consequently rare. The formation has a total thickness of possibly 100 metres and generally exhibits only a slight regional dip of about one degree eastwards (Randal & Brown 1967).

The Jinduckin Formation is a sequence of similarly flat-lying ferruginous quartz sandstone, siltstone, limestone and other shallow water deposits, with a total thickness of about 150 m. Fossils from the upper part of the formation have been identified as Early Ordovician in age, but the lower beds might possibly lie within the Cambrian. (Randal & Brown 1967).

Sampling

The mudstone unit of the Montejinni Limestone was sampled at four widely separated creek exposures, each representing a single horizon within the formation. The 18 hand samples so obtained were oriented using either a Brunton compass or a Sun compass clinometer.

Twenty-eight samples of the Jinduckin Formation were similarly collected at five separate localities, but at one of these (C) a six-metre thick section of the sediments was exposed in two long silage pits which had been excavated from a hillside. Samples here were taken from five different levels, vertically separated by a metre or more, so that the total number of sites sampled in this formation is considered to be nine.

Because a lengthy interval of time may be represented by only a thin layer of sediments, and partly because of their weak and potentially inhomogeneous magnetization, it was decided to cut mostly disc-shaped specimens (3·5 cm × 1·0 cm) from the samples collected. However, such discs cannot at present be accommodated by the tumbling device in our alternating field demagnetization apparatus so some cylindrical specimens (2·2 cm × 2·2 cm) were also cut.

In general, two or more specimens per sample were measured in the laboratory, the totals being 36 from the Montejinni Limestone and 64 from the Jinduckin Formation.

Laboratory measurements

The palaeomagnetic data derived from the two formations are summarized in Table 2. Mean directions have been considered random if the resultant vector (R) is less than the appropriate value tabulated by Watson (1956) or Vincenz & Bruckshaw (1960) which is exceeded in only five per cent of cases by a set of randomly directed vectors. Where site means of only two samples ($N = 2$) are tabulated as non-random, they are associated with vector resultants which exceed 1·93.

The measurements may be grouped into four types.

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Table 2

Site mean directions

Site	Site position Long °E	Lat °S	NRM N(n)	*D*	*I*	*R*	k	Cleaned remanence T	N(n)	*D*	*I*	*R*	k	Site pole Long°	Lat°
Jinduckin Formation															
JA	132·3	14·9	4(6)	359	−58	3·95	62	500	3(6)	251	−23	2·79	9·3	26E	15S
JB	132·2	14·6	3(8)	004	−31	2·99	191	500	4(9)	290	−27	3·63	8·1	33E	22N
JC1	131·2	13·6	2(2)	338	−24	1·96	25	500	2(4)		*	0·92			
JC2	131·2	13·6	4(5)	316	−31	3·72	11	500	3(6)	228	−25	2·93	27	14E	36S
JC3	131·2	13·6	2(4)	349	−26	1·95	20	500	2(5)	260	+15	1·95	19	46E	11S
JC4	131·2	13·6	2(4)	279	−09	1·98	43	500	2(4)	238	+01	1·98	44	34E	31S
JC5	131·2	13·6	2(4)	291	−33	1·95	20	500	2(4)	253	+00	1·94	18	37E	17S
JD	131·3	14·5	4(5)	055	+20	3·85	20	550	4(8)	065	+37	3·93	43	14E	18S
JE	131·6	14·4	2(6)	009	−32	1·98	51	550	2(4)	(003	−20)	1·99	113		
Jinduckin Formation mean direction (α_{95} = 20°)									7	252	−14	6·376	9·6		
Mean pole (α_{95} = 17°)									7			6·571	14	29 °E	16 °S
Montejinni Limestone															
MLA	131·7	15·8	6(11)	016	−50	5·93	70		6(11)						
MLB	131·8	16·1	4(9)	356	−52	3·39	4·9		4(8)						
MLC	131·6	16·9	4(5)	358	−49	3·97	93	600	4(4)	001	−37	3·77	13		
MLD	131·3	17·6	4(10)	349	−44	3·87	24	600	4(8)	209	+53	3·37	4·8		

N = Number of hand samples
n = number of specimens measured
D = Declination in degrees east of north
I = Inclination in degrees, positive downwards
R = Resultant of *N* unit vectors
k = Fisher's precision parameter
T = Temperature (°C) normally used for cleaning
* = directions are random by criteria in text
α_{95}, A_{95} = semi-angles of cones of 95 per cent confidence for mean direction and mean pole respectively

(*a*) *Natural remanent magnetization* (*NRM*)

The direction and intensity of magnetization of specimens from all samples were measured using an astatic magnetometer (Collinson *et al.* 1957). The directional results are listed in Table 2 and it can be seen from Figs 3(a) and 4(a) that most of the 13 site mean directions were close to that of the present geomagnetic field in the area. The intensities of magnetization were fairly weak, ranging from 3·0 to 17×10^{-6} emu cm^{-3} for the Montejinni Limestone and 0·6 to 10×10^{-6} emu cm^{-3} for the Jinduckin Formation.

(*b*) *Alternating field* (*a.f.*) *demagnetization*

Although thermal treatment is commonly applied to sediments, some red beds have been found to respond favourably to magnetic cleaning (e.g. Creer 1959; Jones & McElhinny 1967). It was therefore decided to subject representative cylindrical specimens from both formations to a.f. demagnetization using apparatus similar to that described by McElhinny (1966).

However, even with the maximum available peak field of 2100 oersteds, this type of cleaning failed to remove all the secondary magnetic components present. Progressive thermal demagnetization was therefore applied to most of the remaining specimens.

(*c*) *Thermal demagnetization*

Specimens were heated to successively higher temperatures and cooled in a field-free space, maintained constant to less than 0·5 gamma, using the apparatus

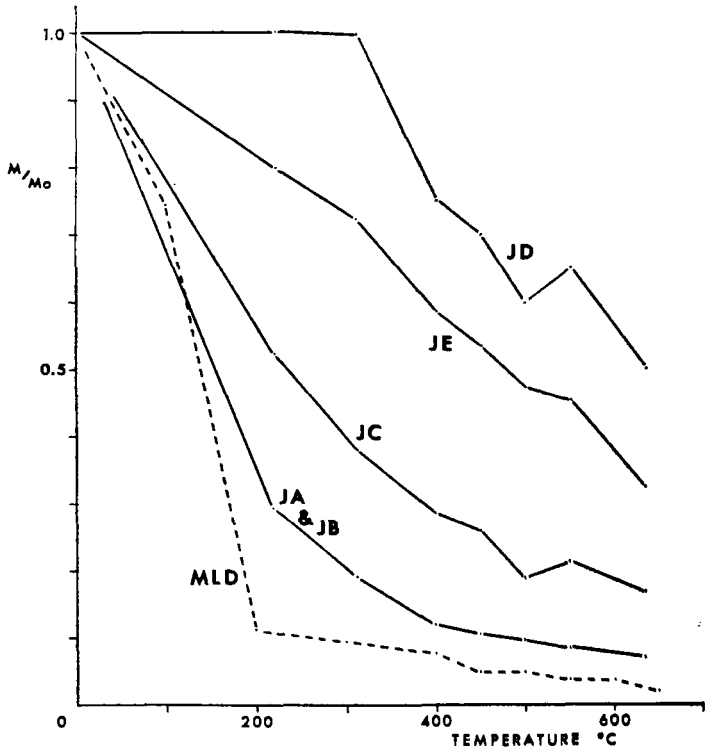


FIG. 2. Thermal demagnetization curves for specimens from the five localities in the Jinduckin Formation (curves for JA and JB are vitually identical) and from one site in the Montejinni Limestone (MLD).

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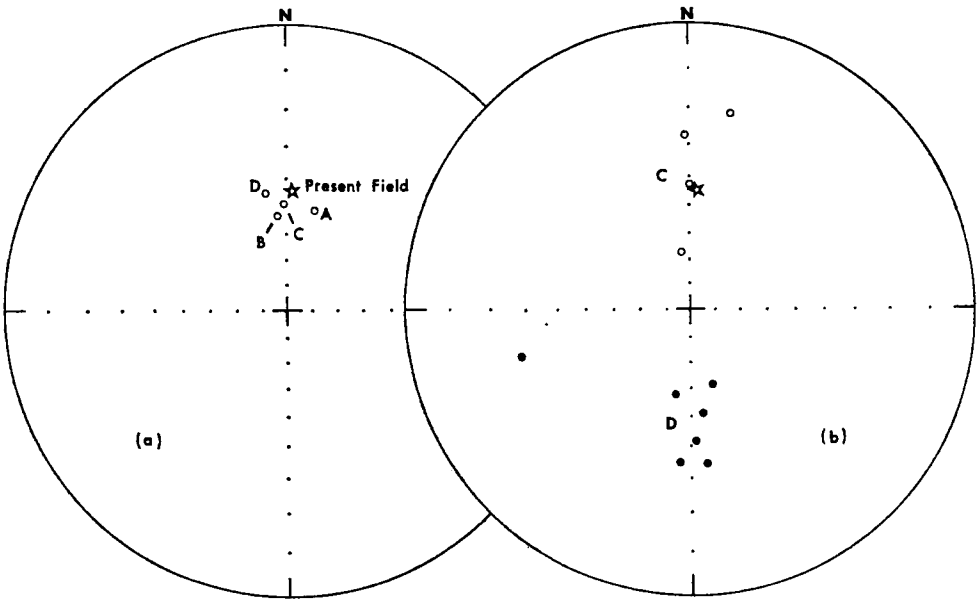


FIG. 3. MONTEJINNI LIMESTONE. Stereographic projections of directions of magnetization, solid symbols on lower hemisphere, open symbols on upper hemisphere. The present geomagnetic field is represented by the star. (a) NRMs of the four sites. (b) Specimens from sites MLC and MLD after demagnetization at a temperature of 600 °C.

described by McElhinny, Luck & Edwards (1970). Typical demagnetization curves are shown in Fig. 2.

Treatment of samples from the Montejinni Limestone produced somewhat diverse effects, the common feature being a rapid decrease in magnetic intensity. This is illustrated by the demagnetization curve for site MLD, which shows that only ten per cent of the NRM value was retained at 200 °C. Sites MLA and MLB were considered palaeomagnetically unacceptable after heating because samples displayed one or more of the following characteristics:

(i) Individual specimens failed to yield a stable direction of magnetization;

(ii) Specimens became magnetically too weak for measurement;

(iii) Specimens remained magnetized along the direction of the present geomagnetic field; and

(iv) The within-sample directions were random.

Site MLC was rejected from consideration because all samples exhibited the third feature described above, although some increase in scatter was observed.

Samples from site MLD attained stable directions of magnetization, all but one of which were almost exactly opposed to that of the present Earth's field (Fig. 3 (b)).

This collection from the Montejinni Limestone has produced no promising results and the formation is therefore regarded as unsuitable material for further palaeomagnetic investigations. In contrast, the Jinduckin Formation responded quite well to thermal cleaning and satisfactory data were obtained from seven of the nine sites.

All samples from sites JA, JB and JD, with the exception of one that became magnetically too weak for measurement, gave stable directions of magnetization and produced reasonably well-grouped site means (Table 2, Fig. 4 (b)).

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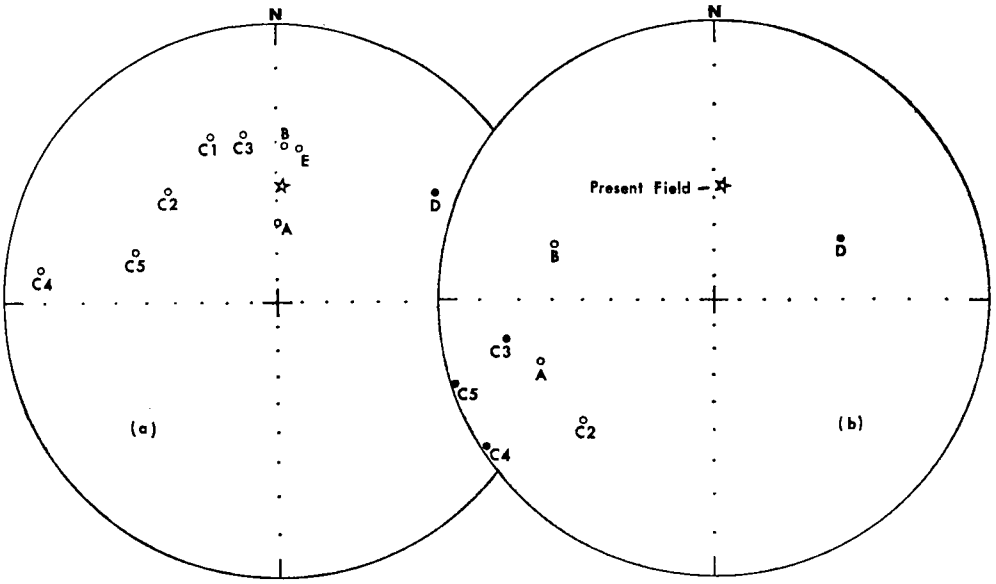


FIG. 4. JINDUCKIN FORMATION. Stereographic projections and symbols as in Fig. 3. (a) NRMs of the nine sites. (b) The seven grouped sites after thermal demagnetization. Site JD is seen to be reversed with respect to the others.

The five sites from locality JC are listed in stratigraphic order in Table 2, JC5 representing the topmost level. Significant mean directions of magnetization after cleaning are shown by sites JC2 to JC5, but the two samples from JC1 have a vector resultant of only 0·92. One of these samples had a direction in approximate agreement with that of site JD, which is reversely magnetized with respect to the other sites, while the second sample tended strongly towards the direction of the others. There is thus the possibility of a reversal within site JC1, a not unreasonable suggestion since two reversals were recently found within a single sample of Cambrian sediments (McElhinny 1970). However, in the present instance, the conservative view was taken and the measurements from JC1 were rejected.

Site JE failed to respond to thermal cleaning and the direction of magnetization of each sample remained close to that of the present geomagnetic field. Because the sandstone from this site was rather coarse and poorly cemented, a chemical demagnetization method (Collinson 1967) was attempted.

(*d*) *Chemical demagnetization*

Two specimens from site JE were immersed in hydrochloric acid for successively increasing intervals of time, after each of which they were briefly washed in distilled water, dried and measured. Little change was noted after immersion in 3 N acid for a total of 17 days. It was then found that a period of 24 hours in 12 N acid partly bleached the specimens, reduced their intensities of magnetization to values close to the minimum measurable, but failed to affect their directions. A chemical demagnetization device such as that described by Burek (1969) might prove more effective, but for the present purpose site JE was rejected from further consideration.

Discussion of results

Although no significant palaeomagnetic results have been obtained from the Montejinni Limestone, the cleaned direction of magnetization from site MLD is of

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some interest. This site differs from the others in that the sediments tend to be coarser-grained, appear rather more weathered and are overlain by a thick surficial layer of Quaternary deposits. This is in contrast to the other sites, which are all overlain by outcropping Tertiary or Cretaceous rocks. It is proposed that the samples from site MLD have been remagnetized by chemical action. associated with a weathering process, during a recent reversed epoch.

A mean direction of primary magnetization for the Jinduckin Formation has been calculated from the seven accepted site means. It takes into account the ' reversed ' polarity of site JD and is shown together with its corresponding mean pole position in Table 2. This represents the first reliable palaeomagnetic result from the lower Palaeozoic of Australia and will be discussed, with results from Cambrian volcanics, in the following paper (McElhinny & Luck 1970).

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